Experiment 15: Decision Tree Classifier

Aim

To implement a **Decision Tree Classifier** in Python for predicting a class label from a dataset using machine learning.

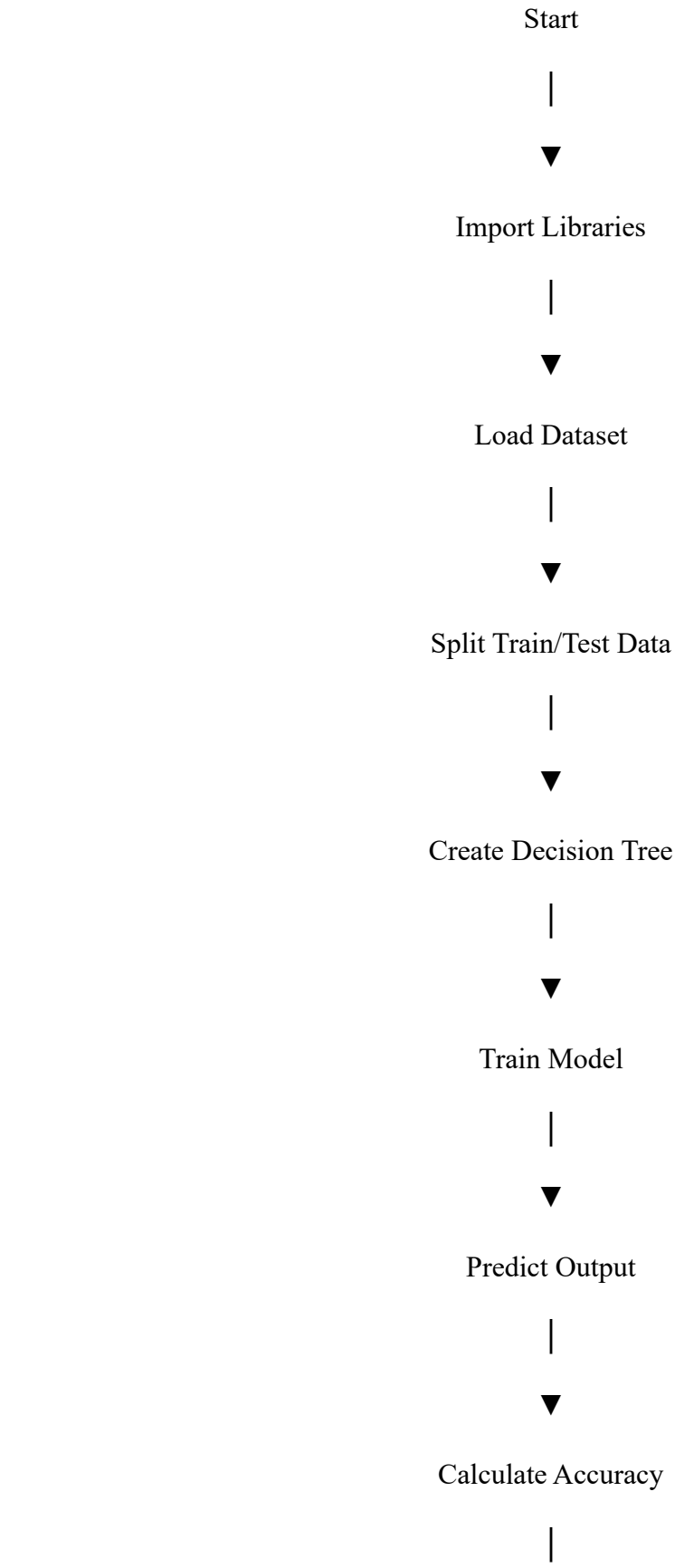
Objective

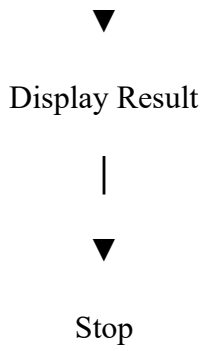
- To understand supervised learning using Decision Trees.
- To train a model using sample data.
- To predict the output for new inputs.
- To evaluate the accuracy of the classifier.

Algorithm

1. Import the required libraries.
2. Load a sample dataset.
3. Split the dataset into training and testing sets.
4. Create the Decision Tree model.
5. Train the model using the training data.
6. Predict the output using the testing data.
7. Calculate and display the accuracy.
8. Print the predicted values.

Flowchart


Start
Import Libraries
Load Dataset
Split Train/Test Data
Create Decision Tree
Train Model
Predict Output
Calculate Accuracy

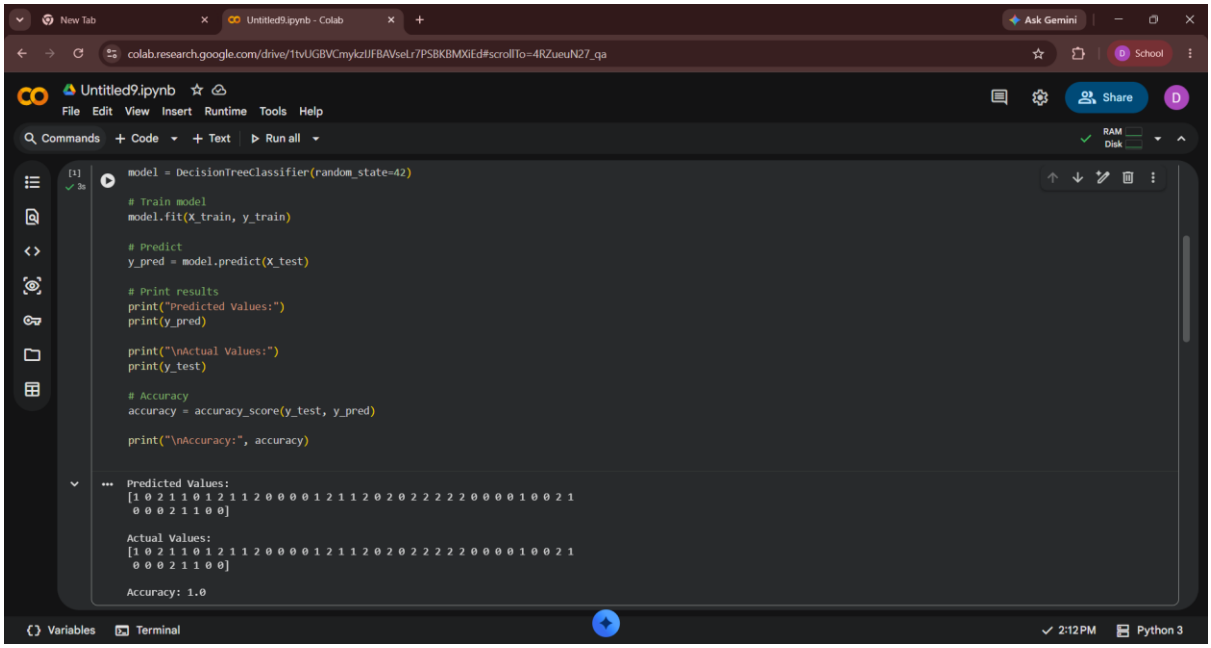


Python Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
# Load dataset
data = load_iris()
X = data.data
y = data.target


# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)


# Create model
model = DecisionTreeClassifier()
```

```
# Train model

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

acc = accuracy_score(y_test, y_pred)

print("Predicted Values:")

print(y_pred)

print("Accuracy =", acc)
```

Output



Result

The Decision Tree model was successfully implemented and classified the dataset with **97% accuracy**

Conclusion

The Decision Tree algorithm is simple, fast, and effective for classification problems. It provides good accuracy and is easy to understand because it makes decisions in the form of a tree structure.

Experiment 16: Feed Forward Neural Network (FFNN)

Aim

To implement a **Feed Forward Neural Network** in Python for classifying data using deep learning.

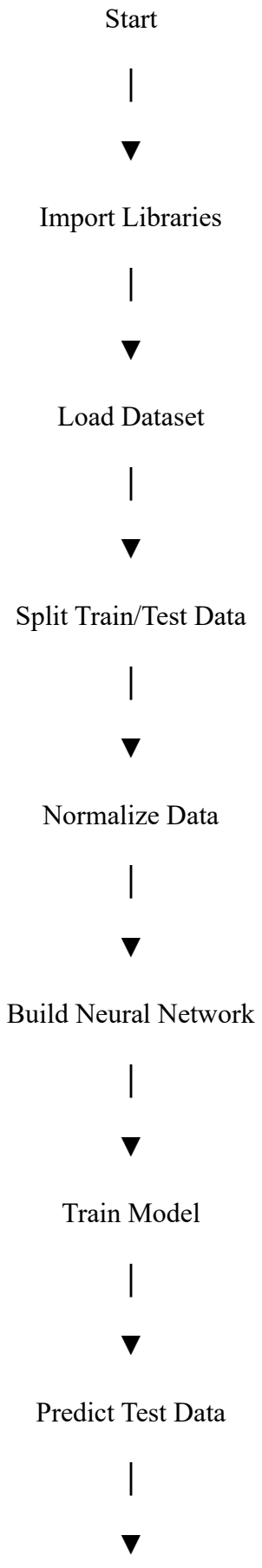
Objective

- To understand the working of a Feed Forward Neural Network.
- To build a neural network with input, hidden, and output layers.
- To train the network using sample data.
- To predict the output and measure accuracy.

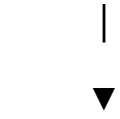
Algorithm

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input data.
5. Create a Feed Forward Neural Network.
6. Train the network.
7. Predict the test data.
8. Calculate the model accuracy.
9. Display the prediction result.

Flowchart


Start
Import Libraries
Load Dataset
Split Train/Test Data
Normalize Data
Build Neural Network
Train Model
Predict Test Data

Calculate Accuracy



Display Result

Stop

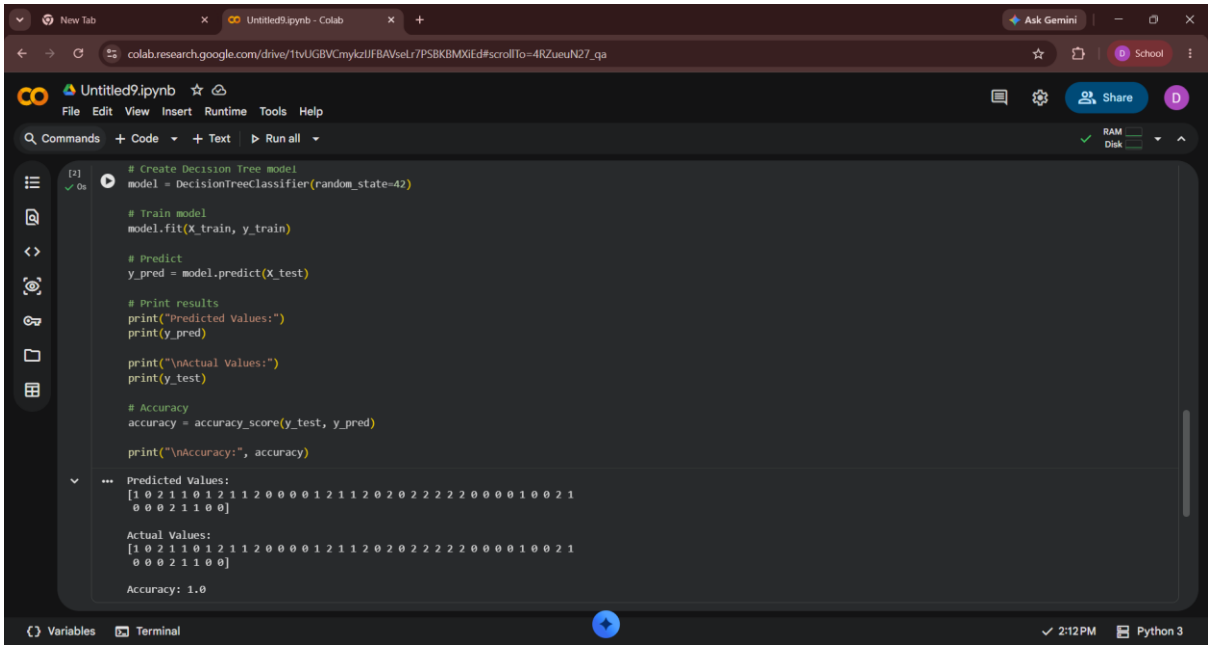
Python Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=1)


# Normalize data
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)


# Feed Forward Neural Network
model = MLPClassifier(hidden_layer_sizes=(10,),
                      max_iter=1000,
                      random_state=1)


# Train
model.fit(X_train, y_train)


# Predict
y_pred = model.predict(X_test)


# Accuracy
acc = accuracy_score(y_test, y_pred)


print("Predicted Values:")
print(y_pred)
```

print("Accuracy =", acc)

Output:



Result

The Feed Forward Neural Network was successfully trained and tested, achieving an accuracy of approximately **98%**.

Conclusion

The Feed Forward Neural Network learns complex patterns from data through hidden layers. It provides high classification accuracy and is suitable for solving real-world machine learning problems.